

ABSOLUTE COLIMITS

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Dedicated to Bob Paré with gratitude for so many beautiful theorems

ABSTRACT. In the context of enriched category theory, we give necessary and sufficient conditions for a module morphism $\alpha: M \rightarrow \mathcal{C}(F, Z)$ to exhibit a functor $Z: \mathcal{A} \rightarrow \mathcal{C}$ as an absolute M -weighted colimit of a functor $F: \mathcal{B} \rightarrow \mathcal{C}$. We also review, with short proofs, the various criteria for all colimits weighted by a particular M to be absolute.

INTRODUCTION

In [15], Paré characterized conical limits of functors $F: \mathcal{B} \rightarrow \mathcal{C}$ which are *absolute* in the sense that they are preserved by all functors out of $\mathcal{C}$. He also characterized conical limits of functors $F: \mathcal{B} \rightarrow \mathcal{C}$ with $\mathcal{C}$ additive which are *additively absolute*, that is, preserved by all additive functors out of $\mathcal{C}$.

In the general enriched context, the second author [20] characterized those weights M for which every existing M -weighted colimit is absolute in the sense that it is preserved by all (enriched) functors. The condition is that M should have a right adjoint module M^* . Given a module M with right adjoint M^* , the first author in [7] showed the equivalence of having an M -weighted colimit for a functor F , having an M^* -weighted limit for F , and having an M -weighted cocone and an M^* -weighted cone making two diagrams commute. To some extent, this was an extended blend of [15] and [20].

In this article, we present a new theorem which is closer to a pure enriched version of [15], but is in fact strong enough to deduce the results in [7, 20].

1. GIVEN WEIGHT AND GIVEN FUNCTOR

We work in the context of enriched categories and weighted (sometimes called “indexed”) colimits; see [21, 10, 12, 18, 19]. The “presheaf” or “right module” category of a category $\mathcal{C}$ will be denoted by $\mathcal{P}\mathcal{C}$ with Yoneda embedding functor $Y: \mathcal{C} \rightarrow \mathcal{P}\mathcal{C}$. We also have $\mathcal{P}^\dagger \mathcal{D} := \mathcal{P}(\mathcal{D}^{\text{op}})^{\text{op}}$ with corresponding Yoneda embedding functor $Y^\dagger: \mathcal{D} \rightarrow \mathcal{P}^\dagger \mathcal{D}$; this is such that

$$(1.1) \quad [\mathcal{D}, \mathcal{P}\mathcal{C}] \cong [\mathcal{C}, \mathcal{P}^\dagger \mathcal{D}]^{\text{op}}.$$

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We write $\mathbf{Mod}$ for the bicategory whose objects are (enriched) categories and whose morphisms are modules, also known as “bimodules”, “profunctors” or “distributors”. A module $M: \mathcal{A} \rightarrow \mathcal{B}$ is a functor $M: \mathcal{A} \rightarrow \mathcal{PB}$ but we write $M(B, A)$ instead of $(MA)B$. The identity module of $\mathcal{B}$, also denoted by $\mathcal{B}$, amounts to the Yoneda embedding.

We are assuming the base for enrichment is rich enough that $\mathbf{Mod}$ should admit right liftings and right extensions [21, 19]. For modules $M: \mathcal{A} \rightarrow \mathcal{B}$ and $K: \mathcal{C} \rightarrow \mathcal{B}$, we write $\mathcal{B}(M, K): \mathcal{C} \rightarrow \mathcal{A}$ for the right lifting of K through M ; it is given on objects by

$$(1.2) \quad \mathcal{B}(M, K)(A, C) = \mathcal{PB}(M(-, A), K(-, C)) .$$

We identify a functor $F: \mathcal{B} \rightarrow \mathcal{C}$ with the module $F: \mathcal{B} \rightarrow \mathcal{C}$ given on objects by $F(C, B) = \mathcal{C}(C, FB)$. With this notation and that of (1.2), we have that $F = \mathcal{C}(1, F): \mathcal{B} \rightarrow \mathcal{C}$. On the other hand, we also have $\mathcal{C}(F, 1): \mathcal{C} \rightarrow \mathcal{B}$, which is right adjoint to $\mathcal{C}(1, F)$ in the bicategory $\mathbf{Mod}$.

For a module $M: \mathcal{A} \rightarrow \mathcal{B}$, the M -weighted colimit of a functor $F: \mathcal{B} \rightarrow \mathcal{C}$, when it exists, is a functor $Z: \mathcal{A} \rightarrow \mathcal{C}$ together with an isomorphism

$$(1.3) \quad \mathcal{C}(Z, 1) \cong \mathcal{B}(M, \mathcal{C}(F, 1)) .$$

By the Yoneda Lemma, such an isomorphism is induced by a module morphism $\alpha: M \Rightarrow \mathcal{C}(F, Z)$ which is said to *exhibit* the colimit. The colimit is *preserved* by a functor $H: \mathcal{C} \rightarrow \mathcal{X}$ when the module morphism

$$\begin{aligned} \mathcal{X}(HZ, 1) &\xrightarrow{\mathcal{X}(HF, -)} \mathcal{B}(\mathcal{X}(HF, HZ), \mathcal{X}(HF, 1)) \xrightarrow{\mathcal{B}(H_F, Z, 1)} \\ &\mathcal{B}(\mathcal{C}(F, Z), \mathcal{X}(HF, 1)) \xrightarrow{\mathcal{B}(\alpha, 1)} \mathcal{B}(M, \mathcal{X}(HF, 1)) \end{aligned}$$

is invertible (the case $H = 1_{\mathcal{C}}$ shows how to recover the isomorphism (1.3) from α). The M -weighted colimit Z is called *absolute* when it is preserved by all functors out of $\mathcal{C}$.

Theorem 1.1. *Consider a module $M: \mathcal{A} \rightarrow \mathcal{B}$ and a functor $F: \mathcal{B} \rightarrow \mathcal{C}$. The following conditions are equivalent:*

- (i) F has an absolute M -weighted colimit;
- (ii) There is a functor $Z: \mathcal{A} \rightarrow \mathcal{C}$ and module morphisms $\alpha: M \Rightarrow \mathcal{C}(F, Z)$ and $\beta: Z \Rightarrow F \circ M$ satisfying (1.4) and (1.5) below;
- (iii) The composite module $F \circ M: \mathcal{A} \rightarrow \mathcal{C}$ is representable (that is, $F \circ M \cong Z$ for some functor $Z: \mathcal{A} \rightarrow \mathcal{C}$);
- (iv) (If idempotents split in $\mathcal{C}$) There exist a functor $Z: \mathcal{A} \rightarrow \mathcal{C}$ and module morphisms $\alpha: M \Rightarrow \mathcal{C}(F, Z)$, $\beta: Z \Rightarrow F \circ M$ satisfying condition (1.4) below.

$$(1.4) \quad \begin{array}{ccc} \mathcal{A} & \xrightarrow{M} & \mathcal{B} \\ & \searrow & \downarrow \alpha \\ & Z & \uparrow \mathcal{C}(F, 1) \\ & \downarrow \beta & \\ \mathcal{B} & \xrightarrow{F} & \mathcal{C} \end{array} \quad = \quad \begin{array}{ccc} \mathcal{A} & \xrightarrow{M} & \mathcal{B} \\ & \searrow \mathcal{B} & \uparrow \mathcal{C}(F, 1) \\ & \eta_F & \\ \mathcal{B} & \xrightarrow{F} & \mathcal{C} \end{array}$$

$$(1.5) \quad \begin{array}{ccc} \mathcal{A} & \xrightarrow{M} & \mathcal{B} \\ \downarrow \beta & \nearrow F & \downarrow \alpha \\ \mathcal{C} & \xrightarrow{\varepsilon_F} & \mathcal{C} \end{array} \quad \begin{array}{c} \nearrow Z \\ \searrow Z \end{array} \quad \begin{array}{c} \mathcal{C} \\ \uparrow \mathcal{C} \end{array} = \begin{array}{ccc} \mathcal{A} & \xrightarrow{M} & \mathcal{B} \\ \downarrow \beta & \nearrow F & \downarrow \alpha \\ \mathcal{C} & \xrightarrow{\varepsilon_F} & \mathcal{C} \end{array} \quad \begin{array}{c} \nearrow Z \\ \searrow Z \end{array} \quad \begin{array}{c} \mathcal{C} \\ \uparrow \mathcal{C} \end{array}$$

Proof. (i) $\Rightarrow$ (ii): Suppose the module morphism $\alpha: M \Rightarrow \mathcal{C}(F, Z)$ exhibits $Z: \mathcal{A} \rightarrow \mathcal{C}$ as an absolute M -weighted colimit of F . Form the collage

$$(1.6) \quad \begin{array}{ccccc} \mathcal{A} & \xrightarrow{M} & \mathcal{B} & \xrightarrow{F} & \mathcal{C} \\ & \searrow U & \swarrow \gamma & \searrow V & \\ & & \mathcal{C}' & & \end{array}$$

of the module $F \circ M$. Herein, U and V are functors and $F \circ M \cong \mathcal{C}'(V, U)$ (cf. [18]). The mate of $\gamma: VFM \Rightarrow U$ under the adjunction $VF \dashv \mathcal{C}'(VF, 1)$ gives a module morphism $\bar{\gamma}: M \Rightarrow \mathcal{C}'(VF, 1) \circ U \cong \mathcal{C}'(VF, U)$, while absoluteness gives $V \circ Z$ as an M -weighted colimit of $V \circ F$ exhibited by

$$(1.7) \quad M \xrightarrow{\alpha} \mathcal{C}(F, Z) \xrightarrow{V} \mathcal{C}'(VF, VZ) .$$

So, by the existence clause in the definition of colimit, there is a module morphism $\beta: \mathcal{A} \Rightarrow \mathcal{C}'(VZ, U) \cong \mathcal{C}'(Z, 1) \circ \mathcal{C}'(V, U) \cong \mathcal{C}'(Z, 1) \circ FM$, whose mate $\bar{\beta}$ under the adjunction $Z \dashv \mathcal{C}'(Z, 1)$ is a module map $Z \Rightarrow F \circ M$. From the fact that $\bar{\beta}$ factors $\bar{\gamma}$ through (1.7), it follows that this β satisfies (1.4); and now (1.5) follows using the uniqueness clause in the definition of colimit.

(ii) $\Rightarrow$ (iii): Let $\bar{\alpha}: F \circ M \Rightarrow Z$ be the mate of α under the adjunction $F \dashv \mathcal{C}(F, 1)$. Clearly, (1.5) says that β and $\bar{\alpha}$ exhibit Z as a retract of $F \circ M$. Now by a simple computation, or a lovely argument using strings, (1.4) says that the idempotent on $F \circ M$ coming from the retraction is the identity.

(iv) $\Rightarrow$ (ii): Much as in the last case, we still obtain an idempotent on Z using (1.4). Split the idempotent on Z to obtain Z', α', β' as in condition (ii).

(ii) $\Rightarrow$ (iv) is obvious.

(iii) $\Rightarrow$ (i): Clearly condition (iii) is preserved by any functor out of $\mathcal{C}$ so all that remains is to see that it implies Z is an M -weighted colimit of F . However, since $F \circ M \cong Z$ by (iii), we have

$$\mathcal{C}(Z, 1) \cong \mathcal{C}(F \circ M, 1) \cong \mathcal{B}(M, \mathcal{C}(F, 1)) ,$$

as in (1.3). $\square$

To obtain the result for limits rather than colimits, notice that the limit of the functor $F: \mathcal{B} \rightarrow \mathcal{C}$ weighted by a module $N: \mathcal{B} \rightarrow \mathcal{A}$ is the colimit of $F: \mathcal{B}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$ weighted by $N^{\text{op}}: \mathcal{A}^{\text{op}} \rightarrow \mathcal{B}^{\text{op}}$.

2. GIVEN WEIGHT AND ALL FUNCTORS

The second author showed in [20] that, when a module $M: \mathcal{A} \rightarrow \mathcal{B}$ has a right adjoint N , all M -weighted colimits are necessarily absolute. In [7], the first author refined this by showing that, in this situation, making a functor $F: \mathcal{B} \rightarrow \mathcal{C}$ into an M -weighted colimit for F is equivalent to making it into an N -weighted limit, and that both are in turn the same as providing:

- A module morphism $\alpha: M \Rightarrow \mathcal{C}(F, Z)$; and
- A module morphism $\bar{\beta}: N \Rightarrow \mathcal{C}(Z, F)$,

subject to the following two conditions:

$$(2.8) \quad \begin{array}{ccc} \mathcal{A} & \xrightarrow{M} & \mathcal{B} \\ \uparrow N & \swarrow \mathcal{C}(Z,1) & \downarrow \alpha \\ \mathcal{B} & \xrightarrow{F} & \mathcal{C} \end{array} \quad \begin{array}{c} \downarrow \varepsilon_Z \\ \mathcal{C} \end{array} \quad \begin{array}{c} \uparrow \mathcal{C}(F,1) \\ \mathcal{C} \end{array} = \begin{array}{ccc} \mathcal{A} & \xrightarrow{M} & \mathcal{B} \\ \downarrow \varepsilon & \swarrow \mathcal{C}(Z,1) & \downarrow \eta_F \\ \mathcal{B} & \xrightarrow{F} & \mathcal{C} \end{array}$$

$$(2.9) \quad \begin{array}{ccc} \mathcal{A} & \xleftarrow{\mathcal{C}(Z,1)} & \mathcal{C} \\ \uparrow \eta & \swarrow N & \downarrow \bar{\beta} \\ \mathcal{A} & \xrightarrow{M} & \mathcal{B} \end{array} \quad \begin{array}{c} \downarrow \alpha \\ \mathcal{C} \end{array} \quad \begin{array}{c} \uparrow \varepsilon_F \\ \mathcal{C} \end{array} = \begin{array}{ccc} \mathcal{A} & \xleftarrow{\mathcal{C}(Z,1)} & \mathcal{C} \\ \uparrow \eta_Z & \swarrow & \downarrow \\ \mathcal{A} & \xrightarrow{M} & \mathcal{B} \end{array}$$

This result is in fact an easy consequence of our Theorem 1.1. Indeed, given α and $\bar{\beta}$ as above, taking the mate of $\bar{\beta}: N \rightarrow \mathcal{C}(Z, F)$ under the adjunctions $Z \dashv \mathcal{C}(Z, 1)$ and $M \dashv N$ yields a module morphism $\beta: Z \Rightarrow F \circ M$, with the conditions (2.8) and (2.9) transforming under the mates correspondence to the conditions (1.4) and (1.5). So in this situation, α exhibits Z as the M -weighted colimit of F , and this colimit is necessarily absolute. Because the data above are self-dual (that is, correspond to an instance of the same data in $\mathcal{C}^{\text{op}}$), it likewise follows that $\bar{\beta}$ exhibits Z as the N -weighted limit of F , and this limit is necessarily absolute.

Note that this argument not only reconstructs the main result of the first author's [7], but also the main result of the second author's [20], namely that colimits by left adjoint modules are necessarily absolute. For completeness, we now provide a result which collates various known characterisations of absolute weights; these can be extracted from Theorem I.4.1 of Dubuc [5], Theoreme I.1.1 of Gouze-Grunig [8], Proposition 1 of Johnson's [9], Chapter 4 Section 1 Exercise 4 (Paré) of Mac Lane [14], and the aforementioned main result of [20].

Theorem 2.1. *The following conditions on a module $M: \mathcal{A} \rightarrow \mathcal{B}$ are equivalent:*

- (i) M is a left adjoint in Mod ;
- (ii) Every M -weighted colimit is absolute;
- (iii) The M -weighted colimit of $Y^\dagger: \mathcal{B} \rightarrow \mathcal{P}^\dagger \mathcal{B}$ is absolute;

- (iv) If $R: \mathcal{B} \rightarrow \mathcal{A}$ is the right lifting of the identity module of $\mathcal{B}$ along M exhibited by the 2-cell $\varepsilon: M \circ R \Rightarrow \mathcal{B}(1, 1)$, then there exists a module morphism $\eta: \mathcal{A}(1, 1) \Rightarrow R \circ M$ such that $(\varepsilon \circ M)(M \circ \eta) = 1_M$;
- (v) For each object U of the base bicategory and each object $A: U \rightarrow \mathcal{A}$ of $\mathcal{A}$, the module $M(-, A): U \rightarrow \mathcal{B}$ has a right adjoint module.

Proof. (i) $\Rightarrow$ (ii): Assume $M \dashv N$ and that $Z: \mathcal{A} \rightarrow \mathcal{C}$ is the M -weighted colimit of $F: \mathcal{B} \rightarrow \mathcal{C}$. Using (1.3), we see that $\mathcal{C}(Z, 1) \cong \mathcal{A}(1, N \circ F) \cong N \circ \mathcal{C}(F, 1)$. Taking left adjoints yields $Z \cong F \circ M$ and the result follows by Theorem 1.1(iii).

(ii) $\Rightarrow$ (iii) is obvious.

(iii) $\Rightarrow$ (i): Let $Z: \mathcal{A} \rightarrow \mathcal{P}^\dagger \mathcal{B}$ be the M -weighted colimit of $Y^\dagger$, and let $N: \mathcal{B} \rightarrow \mathcal{A}$ be the module corresponding to the functor Z under the isomorphism (1.1); the formula is $N(A, B) = \lim(M(-, A), \mathcal{B}(1, B))$. We shall show that M is a left adjoint for N , or equivalently, that M is the absolute right extension of $\mathcal{B}: \mathcal{B} \rightarrow \mathcal{B}$ along N . In other words, we must show that, for all modules $T: \mathcal{B} \rightarrow \mathcal{X}$, the module $T \circ M$ is the right extension of T along N . Notice that, using Yoneda, we have

$$(T \circ M)(-, A) \cong \text{colim}(M(-, A), \mathcal{P}^\dagger \mathcal{B}(T, Y^\dagger)) \cong (Y^\dagger \circ M)(T, A).$$

By Theorem 1.1(iii), $Y^\dagger \circ M \cong Z$. However, $Z(T, A) \cong \lim(N(A, -), T)$ showing that $Z(T, -) \cong T \circ M$ is the right extension of T along N .

(i) $\Leftrightarrow$ (iv): The $\Rightarrow$ is obvious. In the other direction, the data of (iv) are a candidate for unit and counit for R as right adjoint to M . However, only one of the adjunction identities is assumed to hold. By Paré's Exercise in [14], the right adjoint to M is obtained by splitting an idempotent on R .

(i) $\Leftrightarrow$ (v): The $\Rightarrow$ is again obvious. In the other direction, let $\mathcal{U}$ be the category whose objects are those of $\mathcal{A}$ and whose homs $\mathcal{U}(A, B)$ are all initial objects unless $A = B$ in which case the hom is an identity morphism in the base bicategory. So the inclusion $J: \mathcal{U} \rightarrow \mathcal{A}$ is opmonadic (= "of Kleisli type") in Mod ; see [18]. By a dual to Dubuc's Adjoint Triangle Theorem [4], it follows that M has a right adjoint if $M \circ J$ does, and the result follows. $\square$

3. EXAMPLES

Example 3.1. Let $\mathcal{V} = \text{Set}$, let $\mathcal{A} = \mathbf{1}$ (the terminal category), let $\mathcal{B}$ be the free category on the graph with two vertices $0, 1$ and two edges $1 \rightrightarrows 0$, and let $M: \mathbf{1} \rightarrow \mathcal{B}$ be the terminal module. In this case, if $F: \mathcal{B} \rightarrow \mathcal{C}$ takes the generating graph to the parallel pair $u, v: X \rightrightarrows Y$ in $\mathcal{C}$, then a module morphism $\alpha: M \Rightarrow \mathcal{C}(F, Z)$ involves maps $\alpha_1: X \rightarrow Z$ and $\alpha_0: Y \rightarrow Z$ with $\alpha_0 u = \alpha_1 = \alpha_0 v$. Putting $a = \alpha_0$, this is just a map $a: Y \rightarrow Z$ with $au = av$, i.e., a cofork under (u, v) : so an M -weighted colimit of F is a coequaliser of u and v .

Now look at Theorem 1.1(ii). The module morphism $\alpha: M \Rightarrow \mathcal{C}(F, Z)$ is a cofork $a: Y \rightarrow Z$ under (u, v) , and by Yoneda, the module morphism $\beta: Z \Rightarrow F \circ M$ amounts to an element $\beta \in (F \circ M)(Z) = \text{colim}_{B \in \mathcal{B}} \mathcal{C}(Z, FB)$. Now (1.5) says that β has a representative $b \in \mathcal{C}(Z, Y)$ with $ab = 1_Z$; while (1.4) is the requirement that ba and 1_Y are identified in the coequalizer of the parallel pair $\mathcal{C}(Y, u), \mathcal{C}(Y, v): \mathcal{C}(Y, X) \rightrightarrows \mathcal{C}(Y, Y)$ in Set . By the usual construction of

coequalisers in $\mathbf{Set}$, we thus re-find Proposition 5.3 of [15], characterising the data of an absolute coequaliser as being:

- A map $a: Y \rightarrow Z$ with $au = av$;
- A map $b: Z \rightarrow Y$ with $ab = 1_Z$;
- Maps $y_0, \dots, y_n: Y \rightarrow Y$ and $x_1, \dots, x_n: Y \rightarrow X$ with $y_0 = 1_Y$ and $y_n = ba$ and, for each $1 \leq i \leq n$, either $ux_i = y_{i-1}$ and $vx_i = y_i$, or $ux_i = y_i$ and $vx_i = y_{i-1}$.

Example 3.2. Take $\mathcal{V} = \mathbf{Cat}$, so that $\mathcal{V}$ -categories are 2-categories. As in [16], we have the 2-category $\mathbf{Adj}$ freely generated by objects X and Y and an adjunction $f \dashv u: X \rightarrow Y$. Let $\mathcal{A} = \mathbf{1}$, the terminal 2-category, let $\mathcal{B}$ be the full sub-2-category $I: \mathcal{B} \hookrightarrow \mathbf{Adj}$ of $\mathbf{Adj}$ on the object Y , and let $M = \mathbf{Adj}(I, X): \mathbf{1} \dashv \mathbf{B}$. In this case $\mathcal{B}$ is the suspension of Δ_+ , the free strict monoidal category containing a monoid, so that a $\mathcal{V}$ -functor $F: \mathcal{B} \rightarrow \mathcal{C}$ is a monad (T, t) in $\mathcal{C}$. A module morphism $\alpha: M \Rightarrow \mathcal{C}(F, Z)$ is now given by a 1-cell $a: T \rightarrow Z$ together with a 2-cell $\varphi: at \Rightarrow a$ satisfying associativity and unitality laws; we may speak of a *right t -module*. An M -weighted colimit for F is a universal right t -module, so a *Kleisli object* for (T, t) ; see [17].

Consider Theorem 1.1(ii). Now α is a right t -module $(a: T \rightarrow Z, \varphi: at \Rightarrow a)$, while by Yoneda, $\beta: Z \Rightarrow F \circ M$ is an object of $(F \circ M)(Z) = \mathcal{C}(Z, T)_{\mathcal{C}(Z, t)}$, the Kleisli category in $\mathbf{Cat}$ of the monad $\mathcal{C}(Z, t): \mathcal{C}(Z, T) \rightarrow \mathcal{C}(Z, T)$, so equally an object $b \in \mathcal{C}(Z, T)$. Now, condition (1.5) tells us like before that $ab = 1_Z$; while condition (1.4) makes the two assertions that:

- $ba = 1_T$ in $\mathcal{C}(T, T)_{\mathcal{C}(T, t)}$, that is, b and a are inverse isomorphisms;
- The image of the map $ba\varphi: bat \Rightarrow ba$ of $\mathcal{C}(T, T)$ in $\mathcal{C}(T, T)_{\mathcal{C}(T, t)}$ is equal to the universal map $bat \Rightarrow ba$, which is to say that

$$1_{bat} = bat \xRightarrow{b\varphi} ba \xRightarrow{ba\eta} bat .$$

But since a is a right t -module, we also have that $b\varphi \circ ba\eta = 1_{ba}$. Thus $ba\eta = \eta$ is an invertible 2-cell, and so the monad (T, t) must be trivial, that is, isomorphic to the identity monad on T . Thus, a Kleisli object is absolute in a 2-category if and only if it is the Kleisli object of a trivial monad.

Remark 3.3. The preceding result is rather like the situation with coproducts in ordinary categories: a coproduct is absolute if and only if it is a trivial—which is to say, unary—coproduct. Of course, this result is sensitive to the enriching base: in a category enriched over commutative monoids, all finite coproducts are absolute, while in a category enriched over complete join-lattices, *all* coproducts are absolute. In a similar way, the situation with Kleisli objects changes completely if we consider enrichment over categories with reflexive coequalisers: in that context, *every* Kleisli object is absolute [18].

Example 3.4. Take $\mathcal{V} = \mathbf{Cat}$, take $\mathcal{A} = \mathbf{1}$ as before, and take $\mathcal{B}$ to be the locally discrete 2-category to the left in

$$2 \begin{array}{c} \xrightarrow{p} \\ \xleftarrow{-m-} \\ \xrightarrow{q} \end{array} 1 \begin{array}{c} \xrightarrow{d} \\ \xleftarrow{-i-} \\ \xrightarrow{c} \end{array} 0 \qquad W \begin{array}{c} \xrightarrow{p} \\ \xleftarrow{-m-} \\ \xrightarrow{q} \end{array} X \begin{array}{c} \xrightarrow{d} \\ \xleftarrow{-i-} \\ \xrightarrow{c} \end{array} Y \qquad 3 \begin{array}{c} \xleftarrow{01} \\ \xleftarrow{-02-} \\ \xleftarrow{12} \end{array} 2 \begin{array}{c} \xleftarrow{0} \\ \xleftarrow{-0-} \\ \xleftarrow{1} \end{array} 1$$

subject to the simplicial identities $di = ci = 1_0$, $dp = dm$, $cm = cq$ and $cp = dq$. We denote the image of a 2-functor $F: \mathcal{B} \rightarrow \mathcal{C}$ as centre above. We consider $M: \mathbf{1} \rightarrow \mathcal{B}$ the module whose values are displayed right above. Here, **2** and **3** are the posetal categories $0 \leq 1$ and $0 \leq 1 \leq 2$, and the functors are named by giving their images.

A module morphism $\alpha: M \Rightarrow \mathcal{C}(F, Z)$ is a *codescent cocone* under F , whose data are the 1-cell $a = \alpha_0(0): Y \rightarrow Z$ and 2-cell $\theta = \alpha_1(0 \leq 1): f \circ d \Rightarrow f \circ c$, subject to the conditions that $\theta i = 1_f$ and $\theta m = \theta q \circ \theta p$. An M -weighted colimit of F is a universal codescent cocone, also termed a *codescent object* for F .

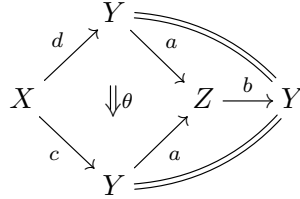
When $\mathcal{C} = \text{Cat}$, the codescent object of a diagram $F: \mathcal{B} \rightarrow \text{Cat}$ is the functor $A_F: Y \rightarrow \text{codesc}(F)$ obtained by adjoining to Y a family of morphisms $(\Theta_x: d(x) \rightarrow c(x))_{x \in X}$ (which will be the components of $\Theta: A \circ d \Rightarrow A \circ c$) subject to the following identities:

- (i) $c(f) \circ \Theta_x = \Theta_{x'} \circ d(f)$ for all $f: x \rightarrow x'$ in X ;
- (ii) $\Theta_{i(y)} = 1_y$ for all $y \in Y$;
- (iii) $\Theta_{m(w)} = \Theta_{q(w)} \circ \Theta_{p(w)}$ for all $w \in W$.

Note, in particular, that A is the identity on objects.

Now, for a general $F: \mathcal{B} \rightarrow \mathcal{C}$, we consider once again Theorem 1.1(ii). The module morphism $\alpha: M \Rightarrow \mathcal{C}(F, Z)$ is a codescent cocone (a, θ) , while $\beta: Z \Rightarrow F \circ M$ is an object $\beta \in (F \circ M)(Z) = \text{codesc}(\mathcal{C}(Z, F-))$, so equally an object of $\mathcal{C}(Z, Y)$. Once again, (1.5) says that $ab = 1_Z$; while (1.4) becomes the two requirements that:

- ba and 1_Y are identified by $A_{\mathcal{C}(Y, F)}: \mathcal{C}(Y, Y) \rightarrow \text{codesc}(\mathcal{C}(Y, F))$. Since this functor is bijective on objects, this just says that $ba = 1_Y$.
- The morphism of $\mathcal{C}(X, Y)$ given by:



is identified by $A_{\mathcal{C}(X, F)}: \mathcal{C}(X, Y) \rightarrow \text{codesc}(\mathcal{C}(X, F))$ with the universally adjoined morphism $\Theta_{\text{id}_X}: d \rightarrow c$.

Of course, the first requirement tells us that a is an isomorphism with inverse b . However, the second requirement is quite non-trivial, due to the interactions of the identities (i)–(iii) for a codescent object in Cat , and a complete characterisation in the spirit of [15] would be both unpleasant and not particularly enlightening. In lieu of this, we provide a few *sufficient* conditions for a codescent cocone to be absolute, which should illuminate the scope of the possibilities.

Since we know already that a is invertible, we may as well assume it to be the identity: thus, our codescent cocone under F is specified by a single 2-cell $\theta: d \Rightarrow c: Y \rightarrow X$ satisfying $\theta i = 1_{1_X}$ and $\theta m = \theta q \circ \theta p$.

- Suppose there is a 2-cell $\gamma: i \circ d \Rightarrow 1_X: X \rightarrow X$ such that $d \circ \gamma = 1_d$ and $c \circ \gamma = \theta$. Then the codescent cocone is absolute, since in $\text{codesc}(\mathcal{C}(Y, F))$

we have that

$$\Theta_{1_X} = \Theta_{1_X} \circ 1_d = \Theta_{1_X} \circ d(\gamma) = c(\gamma) \circ \Theta_{i(d)} = c \circ \gamma = \theta .$$

- Dually, the codescent cocone is absolute if there is $\delta: 1_X \Rightarrow ic$ with $d\delta = \theta$ and $c\delta = 1_c$.
- Suppose there is $k: X \Rightarrow W$ for which $mk = 1_X$, along with 2-cells $\gamma: id \Rightarrow pk: X \rightarrow X$ and $\delta: qk \Rightarrow ic: X \rightarrow X$ for which

$$d\gamma = 1_d \quad c\delta = 1_c \quad \text{and} \quad d\delta \circ c\gamma = \theta .$$

Then the codescent cocone is absolute, since:

$$\begin{aligned} \Theta_{1_X} &= \Theta_{mk} = \Theta_{qk} \circ \Theta_{pk} = c\delta \circ \Theta_{qk} \circ \Theta_{pk} \circ d\gamma \\ &= \Theta_{ic} \circ d\delta \circ c\gamma \circ \Theta_{id} = d\delta \circ c\gamma = \theta . \end{aligned}$$

- Suppose there is a 1-cell $g: X \rightarrow X$ such that $dg = d$ and $cg = c$, along with 2-cells $\gamma: i \circ d \Rightarrow g$ and $\delta: 1_X \Rightarrow g: X \rightarrow X$ such that

$$d\gamma = 1_d \quad c\gamma = \theta \quad d\delta = 1_d \quad \text{and} \quad c\delta = 1_c .$$

Then the codescent cocone is absolute, since:

$$\Theta_{1_X} = c\delta \circ \Theta_{1_X} = \Theta_g \circ d\delta = \Theta_g \circ d\gamma = c\gamma \circ \Theta_{id} = c\gamma = \theta .$$

The reader is encouraged to explore other examples combining the techniques illustrated above.

4. REMARKS ON CAUCHY WEIGHTS

To conclude the paper, we discuss some important necessary conditions to be a Cauchy weight, and circumstances under which these are in fact sufficient. In this section, we assume our base for enrichment $\mathcal{V}$ is a monoidal category (and not a bicategory.)

Proposition 4.1. *Let $\mathcal{V}$ be locally finitely presentable as a closed category [11], and let $\mathcal{C}$ be a $\mathcal{V}$ -category with finitely presentable homs. If the module $M \in [\mathcal{C}^{\text{op}}, \mathcal{V}]$ is finitely presentable—and in particular, if it is Cauchy—then it lands in the subcategory $\mathcal{V}_{\text{fp}}$ of $\mathcal{V}$ consisting of the finitely presentable objects.*

Proof. For such a finitely presentable M and any $A \in \mathcal{C}$, we must show that $[MA, -]: \mathcal{V} \rightarrow \mathcal{V}$ preserves filtered colimits. We have isomorphisms

$$\begin{aligned} [MA, V] &\cong \left[\int^B MB \otimes \mathcal{C}(A, B), V \right] \\ &\cong [\mathcal{C}^{\text{op}}, \mathcal{V}](M, [\mathcal{C}(A, -), V]) \end{aligned}$$

both $\mathcal{V}$ -natural in V . Since each $\mathcal{C}(A, B)$ is finitely presentable, $[\mathcal{C}(A, B), V]$ preserves filtered colimits in the variable V . Since colimits in $[\mathcal{C}^{\text{op}}, \mathcal{V}]$ are formed pointwise, $[\mathcal{C}(A, -), V]$ preserves filtered colimits in the variable V . Then M finitely presentable implies $[\mathcal{C}^{\text{op}}, \mathcal{V}](M, [\mathcal{C}(A, -), V])$ preserves filtered colimits in the variable V . $\square$

In the case that $\mathcal{V}$ is not just locally finitely presentable as a closed category, but “locally 0-presentable”—i.e., $\mathcal{V} = [\mathcal{C}^{\text{op}}, \text{Set}]$ under the convolution monoidal structure coming from a symmetric monoidal structure on $\mathcal{C}$ —an analogous proof shows that: if $\mathcal{C}$ is a $\mathcal{V}$ -category with small-projective homs, then any small-projective (= Cauchy) $M \in [\mathcal{C}^{\text{op}}, \mathcal{V}]$ takes values in the subcategory $\mathcal{V}_{\text{sp}}$ of small-projective objects.

Proposition 4.2. *Let $\mathcal{V}$ be locally finitely presentable as a closed category. Let $\mathcal{C}$ is a small $\mathcal{V}$ -category which can be written as*

$$\mathcal{C} = \bigcup_i \mathcal{C}_i$$

where $\{\mathcal{C}_i : i \in I\}$ is a directed family of full sub- $\mathcal{V}$ -categories, each of which is a cosieve in $\mathcal{C}$, i.e., $\mathcal{C}(W, U) = 0$ for $U \in \mathcal{C}_i$ and $W \notin \mathcal{C}_i$. If $M \in [\mathcal{C}^{\text{op}}, \mathcal{V}]$ is finitely presentable then it is the Kan extension of its own restriction to some $\mathcal{C}_i$. In particular, this is the case if $M : \mathcal{I} \rightarrow \mathcal{C}$ is a Cauchy module.

Proof. Let $M^{(i)} \in [\mathcal{C}^{\text{op}}, \mathcal{V}]$ be the Kan extension of the restriction of M to the full sub- $\mathcal{V}$ -category $\mathcal{C}_i$; thus we have the following, where the second equality follows since $\mathcal{C}_i$ is a cosieve in $\mathcal{C}$:

$$M^{(i)}W = \int^{U \in \mathcal{C}_i} \mathcal{C}(W, U) \otimes MU = \begin{cases} MW & \text{if } W \in \mathcal{C}_i, \\ 0 & \text{otherwise.} \end{cases}$$

Clearly, M is the directed colimit of the $M^{(i)}$'s. Since $[\mathcal{C}^{\text{op}}, \mathcal{V}](M, -)$ preserves filtered colimits, we have

$$[\mathcal{C}^{\text{op}}, \mathcal{V}](M, M) \cong \text{colim}_i [\mathcal{C}^{\text{op}}, \mathcal{V}](M, M^{(i)}) .$$

This is true in $\mathcal{V}$ at the enriched level, but also in the underlying category of $\mathcal{V}$. So each endomorphism of M factors through some $M^{(n)}$. In particular, there exists n such that

$$\begin{array}{ccc} M & \xrightarrow{1_M} & M \\ & \searrow & \nearrow \text{in}_n \\ & M^{(n)} & \end{array}$$

commutes. Of course, in_n has invertible component at any $W \in \mathcal{C}_n$. But for $W \notin \mathcal{C}_n$, we have the diagram

$$\begin{array}{ccccc} & MW & \xrightarrow{1_{MW}} & MW & \\ & \nearrow ! & & \searrow ! & \\ 0 & \xrightarrow{1_0} & 0 & & \end{array}$$

which shows in_n is also invertible there, and so $M^{(n)} \cong M$. $\square$

In some important cases, Propositions 4.1 and 4.2 provide the sufficient conditions for an M to be Cauchy. Compare the next result with Lemma 4 of [1].

Proposition 4.3. *Suppose $\mathcal{V}$ is the monoidal category of vector spaces over a field k of characteristic zero. Let $\mathcal{C} = k\mathcal{G}$ be the free $\mathcal{V}$ -category on a groupoid $\mathcal{G}$ with finite homs. A functor $M: \mathcal{C}^{\text{op}} \rightarrow \mathcal{V}$ is Cauchy as a module $\mathcal{I} \leftrightarrow \mathcal{C}$ if and only if Mx is finite dimensional for all objects x and zero on all but finitely many connected components.*

Proof. It remains to prove the “if” direction. It does no harm to replace $\mathcal{G}$ by an equivalent groupoid which is the disjoint union of groups. On doing so, we note that M is a finite direct sum of objects N of $[\mathcal{C}^{\text{op}}, \mathcal{V}]$ with $Nx = 0$ for all but one x . This reduces the problem to basic facts in the linear representation theory of the single finite group $G_x = \mathcal{G}(x, x)$. Using Maschke’s Theorem in the form that monomorphisms split in $[G_x^{\text{op}}, \mathcal{V}]$, we deduce that every finite-dimensional representation is a direct sum of irreducible representations. Using Maschke’s Theorem in the form that epimorphisms split in $[G_x^{\text{op}}, \mathcal{V}]$, we deduce that every irreducible representation is a retract of the group algebra kG_x with the regular action. However, kG_x is the representable functor in $[G_x^{\text{op}}, \mathcal{V}]$ and so is Cauchy. This implies that our finite-dimensional representation is a finite direct sum of retracts of representables and therefore is Cauchy in $[G_x^{\text{op}}, \mathcal{V}]$ and hence in $[\mathcal{G}^{\text{op}}, \mathcal{V}] \cong [\mathcal{C}^{\text{op}}, \mathcal{V}]$. Yet, M is a finite direct sum of such objects and so Cauchy. $\square$

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